

ii) tanh Activation derivative

$$\sigma = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

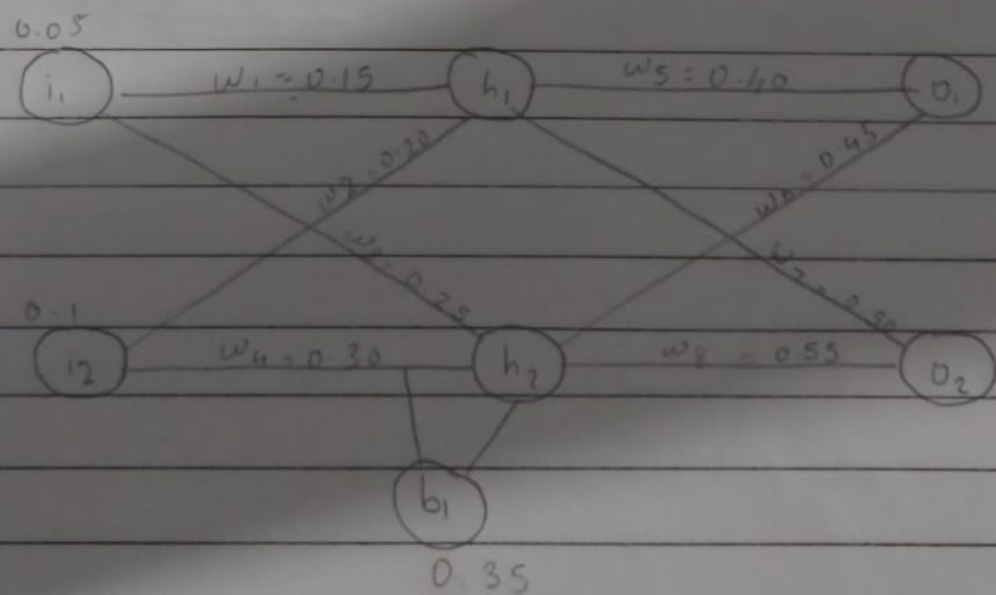
$$\sigma' = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= 1 - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2$$

$$= 1 - \sigma^2$$



Target values $T_1 = 0.01$, $T_2 = 0.99$; $\alpha = 0.$